WireGuard VPN Deployment Guide (Linux Server & Linux Client)

This guide provides step-by-step instructions for deploying a secure WireGuard VPN tunnel between two Linux virtual machines (a VPN Server and a VPN Client). All environment-specific IP addresses, public keys, and sensitive tokens have been sanitized into reusable placeholders.

1. Deployment Environment Parameters

Replace these placeholder values with your own target environment parameters prior to execution:

Parameter	Placeholder / Description	Example Usage
Server Public IP	<SERVER_PUBLIC_IP>	Public IP / FQDN of the WireGuard Server
VPN Subnet CIDR	<VPN_SUBNET_CIDR>	10.0.0.0/24
Server VPN IP	<SERVER_VPN_IP>	10.0.0.1/24
Client VPN IP	<CLIENT_VPN_IP>	10.0.0.2/32
Listen Port	51820	51820/UDP

2. Server-Side Deployment

Step 1: Update Server Packages

```
sudo apt update && sudo apt upgrade -y
```

Step 2: Install WireGuard

```
sudo apt install wireguard -y
```

Verify installation:

```
wg --version
```

Step 3: Create WireGuard Configuration Directory

```
sudo mkdir -p /etc/wireguard
```

Step 4: Generate Server Cryptographic Keys

```
cd /etc/wireguard
wg genkey | sudo tee privatekey | wg pubkey | sudo tee publickey
```

View generated keys and store them securely:

```
sudo cat privatekey
sudo cat publickey
```

Step 5: Create Server Configuration File

Open configuration file:

```
sudo nano /etc/wireguard/wg0.conf
```

Add the following configuration block:

```
[Interface]
Address = <SERVER_VPN_IP>/<SUBNET_MASK>
ListenPort = 51820
PrivateKey = <SERVER_PRIVATE_KEY>
```

Save file (Ctrl + O → Enter → Ctrl + X).

Step 6: Configure Network Security Group (NSG) / Firewall Rule

Ensure inbound UDP traffic is permitted on the WireGuard listening port:

Setting	Value
Source	Any
Protocol	UDP
Destination Port	51820
Action	Allow
Priority	300
Name	Allow-WireGuard

Step 7: Enable and Start WireGuard Service

```
sudo systemctl enable wg-quick@wg0
sudo systemctl start wg-quick@wg0
```

Verify service status:

```
sudo wg show
```

Step 8: Verify Network Interface

```
ip addr show wg0
```

3. Client-Side Deployment

Step 1: Update Client Machine Packages

```
sudo apt update && sudo apt upgrade -y
```

Step 2: Install WireGuard on Client

```
sudo apt install wireguard -y
```

Step 3: Generate Client Cryptographic Keys

```
sudo mkdir -p /etc/wireguard
cd /etc/wireguard
wg genkey | sudo tee privatekey | wg pubkey | sudo tee publickey
```

Step 4: Register Client Peer on Server

Obtain Client Public Key from client machine:

```
sudo cat /etc/wireguard/publickey
```

On the VPN Server, open /etc/wireguard/wg0.conf and append the peer section:

```
[Peer]
PublicKey = <CLIENT_PUBLIC_KEY>
AllowedIPs = <CLIENT_VPN_IP>/32
```

Restart WireGuard on the Server to apply changes:

```
sudo systemctl restart wg-quick@wg0
```

Confirm registration on server:

```
sudo wg show
```

Step 5: Create Client Configuration File

On the Client machine, create /etc/wireguard/wg0.conf:

```
sudo nano /etc/wireguard/wg0.conf
```

Add the following configuration block:

```
[Interface]
PrivateKey = <CLIENT_PRIVATE_KEY>
Address = <CLIENT_VPN_IP>/<SUBNET_MASK>

[Peer]
PublicKey = <SERVER_PUBLIC_KEY>
Endpoint = <SERVER_PUBLIC_IP>:51820
AllowedIPs = <VPN_SUBNET_CIDR>
```

```
PersistentKeepalive = 25
```

Step 6: Start WireGuard Tunnel on Client

```
sudo wg-quick up wg0
```

Verify client status:

```
sudo wg show
```

4. Validate Connectivity & Performance

Handshake Verification

Run on either server or client to verify handshakes:

```
sudo wg show
```

Successful tunnel establishment displays a recent timestamp under latest handshake.

Ping & Packet Loss Test

Execute from Client machine targeting Server VPN IP:

```
ping -c 20 <SERVER_VPN_IP>
```

Latency & RTT Measurement

Check round-trip time statistics from ping summary output:

```
ping -c 20 <SERVER_VPN_IP> | tail -n 2
```

5. Essential Troubleshooting Commands

Task	Command
Check WireGuard status	sudo wg show
Check systemd service status	sudo systemctl status wg-quick@wg0
Check network interface	ip addr show wg0
Verify UDP socket bound	sudo ss -ulpn | grep 51820
View daemon logs	journalctl -u wg-quick@wg0 -f

Deployment Success Checklist

- [ ] WireGuard installed on Server and Client
- [ ] Unique cryptographic key pairs generated for both hosts
- [ ] Firewall / NSG UDP 51820 inbound rule applied on Server
- [ ] Server configuration created with valid Private Key
- [ ] Client configuration created with valid Private Key and Server Endpoint
- [ ] Client Public Key registered under [Peer] on Server
- [ ] Tunnel initialized on both endpoints
- [ ] Handshake successfully established
- [ ] ICMP Echo (Ping) functional across VPN IP addresses